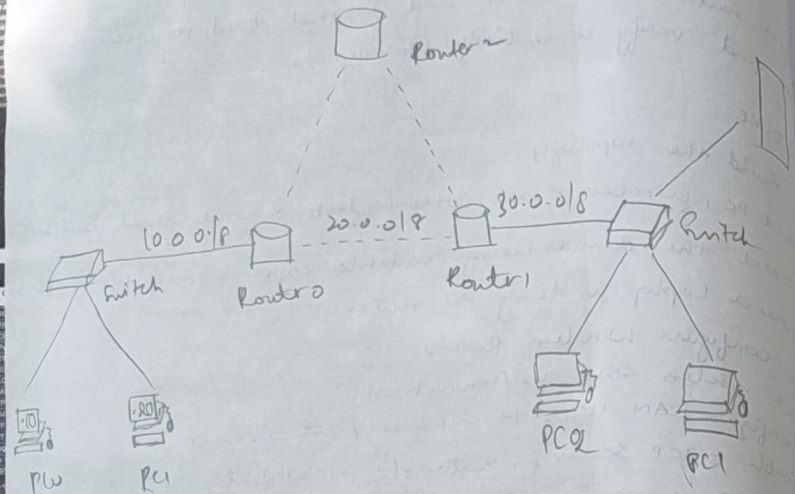
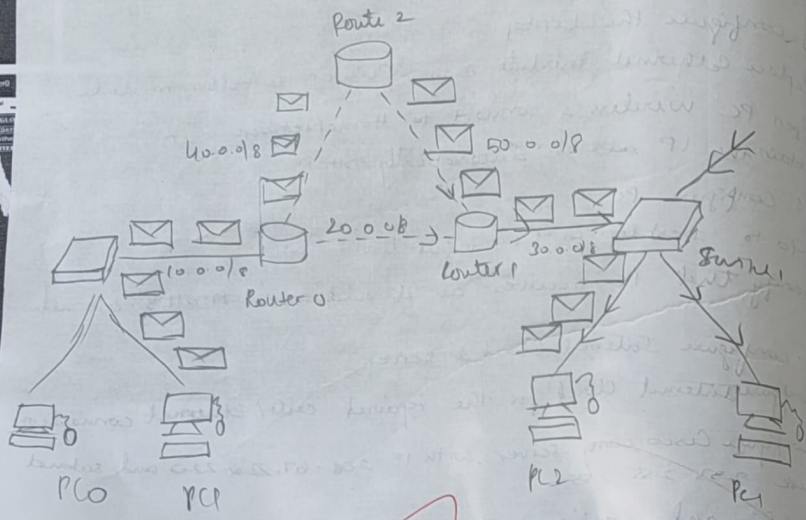




Ex 11-a Static Routing Configuration using Cisco Packet Tracer

AIM: To simulate & configure static Routing b/w multiple networks using Cisco Packet Tracer & verify communication b/w them.

STEPS / PROCEDURE:

1. Create the Network Topology
   - Place 3 Routers, switches, PCs and a server in Packet Tracer
   - Connect the devices as shown in Topology
   - Configure the required IP addresses for all router interfaces.
2. Identify the Networks
   - The networks used are: 10.0.0.0/8, 20.0.0.0/8, 30.0.0.0/8, 40.0.0.0/8 and 50.0.0.0/8
   - Identify the remote networks that are not directly connected to each router.
3. Configure Static Routes
   - Open each router → CLI
   - Enter configuration mode:

     enable
     configure terminal
   - Add routes using

     ip route <destination-network> <subnet-mask> <next-hop>
   - Eg: ip route 30.0.0.0 255.0.0.0 20.0.0.2
4. Configure Main & Backup Routes
   - Configure primary route with lower Administrative Distance (AD)
   - Configure another route with higher AD as backup

     ip route 30.0.0.0 255.0.0.0 20.0.0.2 10
     ip route 30.0.0.0 255.0.0.0 40.0.0.2 20
5. Verify Static Routing:
   - Check configured static routes using:

     show ip route static
   - Test communication using

     ping <destination-IP>
     tracert <destination-IP>
6. Delete a static Route if required: ~~no ip route <destination~~

   no ip route <destination-network> <subnet-mask> <next-hop>

RESULT: Thus static routing with main & backup routes was successfully configured using Cisco Packet Tracer. Connectivity b/w diff networks was verified using ping and tra